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The impact of carbon pricing on a closed-loop supply chain: an Australian case study

Behnam Fahimnia [a,*], Joseph Sarkis [b,1], Farzad Dehghanian [c,2], Nahid Banihashemi [d,3], Shams Rahman [e,4]

[a] University of Technology Sydney, UTS Business School, Management Discipline Group, Haymarket, NSW 2000, Australia
[b] Clark University, Graduate School of Management, Worcester, MA 01610-1477, USA
[c] Ferdowsi University of Mashhad, Department of Industrial Engineering, Faculty of Engineering, Mashhad, Iran
[d] University of South Australia, School of Mathematics and Statistics, Centre for Industrial and Applied Mathematics, Mawson Lakes, SA 5095, Australia
[e] RMIT University, School of Business IT & Logistics, Melbourne, VIC 3000, Australia

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ABSTRACT

Concerns about industrial and supply chain implications on our natural environment have existed for decades. Climate change and greenhouse gas emissions have caused countries to implement various instruments ranging from taxes, permits and voluntary incentives to required regulatory policies. Given this environment, we develop a unified optimization model for a closed-loop supply chain in which the carbon emission is expressed in terms of dollar carbon cost. This study is one of the first to evaluate the forward and reverse supply chain influences on the carbon footprint. A comparative analysis is completed with a decomposition of cost and environmental influences across supply chain functions. We utilize data from a company located in Australia, where the government is currently introducing a carbon pricing scheme. We find that variations in cost and environmental impacts occur over ranges of carbon pricing. Characteristics and patterns of the numerical results over these ranges provide insights for corporate key strategies and potential additional government policies. These results and implications are analyzed along with limitations and directions for future research.

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1. Introduction

As green supply chain management (GSCM) research continues to evolve and progress, dynamic regulatory, social, environmental and industrial concerns provide this research field with many opportunities for further investigation. Regulatory pressures, new industrial competitive requirements and standards, and community expectations all play a role in how industry and its supply chains (SCs) are to be developed (Sarkis et al., 2011; Zhu and Sarkis, 2007). Fledgling investigations in this research field have been relatively descriptive trying to make sense of the practice of GSCM. There have also been nascent efforts on optimization and formal modeling for GSCM. Of over 300 papers focusing on GSCM, just over 10% have focused the use of analytical and optimization modeling (Seuring, 2013). We seek to add to this dimension of research literature, but in a very practically targeted manner.

The importance of GSCM has been duly noted in efforts by national and international organizations to decouple environmental burdens from economic growth as is evidenced by the recent RIO+20 United Nations Conference (Vazquez-Brust and Sarkis, 2012). Part of this green growth effort includes development of closed-loop systems for resource consumption efficiencies in addition to transportation optimization, dual and sometimes complementary goals that industry can utilize. Optimization and efficiencies of GSCM are important contributors to this decoupling. Companies have realized the importance of this decoupling of economic growth and environmental degradation from various external, and sometimes internal, forces (Sarkis, 2012a; Seuring and Müller, 2008; Zhu et al., 2007).

Within this environmental and economically competitive context, the research presented in this paper was motivated by a GSCM planning problem faced by an Australian organization.

* Corresponding author. Tel.: +61 2 95143612; fax: +61 2 95143602.
E-mail addresses: behnam.fahimnia@uts.edu.au (B. Fahimnia), jsarkis@clarku.edu (J. Sarkis), f.dehghanian@um.ac.ir (F. Dehghanian), nahid.banihashemi@unisa.edu.au (N. Banihashemi), shams.rahman@rmit.edu.au (S. Rahman).
[1] Tel.: +1 508 7937659.
[2] Tel.: +98 511 8615100.
[3] Tel.: +61 8 8302 5296.
[4] Tel.: +61 3 9925 5530.

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Utilizing this organizational requirement for a textile manufacturer, we develop a closed-loop SC (CLSC) model that explicitly considers the reduction of environmental pollution (carbon-equivalent emission) in both forward and reverse SC directions. The proposed model uses 'overall SC cost' and 'carbon emissions' as the primary performance measures. The two measures are unified to form a single objective optimization model by expressing the carbon emissions in terms of dollar carbon cost. The model is generalizable enough to be applied in a number of settings. The implications gathered from some parametric analysis provide practical insights with a possibility for generalization and clearly identified future research work. A win–win goal of improving both economic and environmental performance, the core objective of green growth decoupling, is the explicit objective of the CLSC model presented in this paper.

We build on and advance work in optimization of CLSC and GSCM. The balance of economic and environmental perspectives brings the modeling efforts in these two fields together. After a brief introduction into the policy and regulatory issues facing corporations in Australia, this research is placed within the previous literature describing how the work addresses gaps and advances knowledge in this field. We then introduce the model and the case study. The numerical results of executing the bi-criteria model are then presented along with the careful analysis of the results. The implications of the results are the primary focus of the analysis. Further developments and directions for research, as well as limitations are discussed in the Conclusion section.

This paper contributes to the GSCM and CLSC planning literature by focusing on a tactical-operational planning model for carbon fixed price regulatory environments. This study provides a foundation for future research to regulatory and organizational SC settings where carbon taxation (pricing) and carbon market trading are used for industrial carbon emissions management. This contribution objective is met through the development and testing of a mixed integer-linear programming (MILP) formulation of an actual case company in Australia.

2. Background and literature review

2.1. Carbon pricing in Australia and organizational implications

We begin this section by providing a brief historical perspective of Australia and its efforts to manage national carbon emissions. Having the world's highest per capita carbon dioxide emissions (Lo and Spash, 2012), Australia has debated the merits of introducing a carbon or greenhouse gas emissions pricing and trading scheme for over two decades (Nelson et al., 2012). Internally to Australia a number of efforts on emissions trading discussions occurred with various schemes presented and evaluated. Internationally, additional discussions occurred because of the United Nations Framework Convention on Climate Change (UNFCCC), the Kyoto Protocol. Although not signatories to the Kyoto Protocol, the Australian Productivity Commission urged the consideration of market mechanisms, including the use of tradable emission permits, to help the country achieve emissions reduction goals. The Productivity Commission's reports included a debate of whether a determined tax rate on carbon emissions is more effective than free market trading. In 2002 a review considered in significant detail the form and nature of a national emissions trading system (ETS) in Australia with a recommendation of a country wide emissions scheme (Council of Australian Governments, 2002; Nelson et al., 2012).

Various additional Australian studies on the economic implications of carbon pricing, ETS, and economic implications were completed throughout the rest of the 2000's decade. Eventually, due to the Australian government's pledge at the Copenhagen accord, a politically bipartisan supported adoption of ETS policy was agreed upon for commencement in 2012 (Fahimnia et al., 2013; Meng et al., 2013). In 2011 the Commonwealth Government passed the Clean Energy Future package of legislation. An ETS was then established in Australia to take effect from 1 July 2012. The scheme commences with a three year fixed price period with the price in 2012/13 being $23 per tonne of carbon dioxide equivalent. This price will be payable by any corporation operating a facility with annual aggregate emissions more than 25,000 tonnes of carbon equivalent.

The carbon pricing scheme is essentially a carbon tax, a powerful policy mechanism that can help set the market prices for products and energy and to internalize organizational costs of the many external climate change costs (Baranzini et al., 2000). The carbon tax in Australia, according to some simulation results, can cut emissions across the economy effectively, but will cause a mild economic contraction (Meng et al., 2013). Not all economic sectors will necessarily be influenced equally and some sensitive sectors such as soft coal industry are small enough not to have great influence on economic contraction (Fahimnia et al., 2013; Meng, 2012).

The policy theory behind reducing emissions through market-based mechanisms (carbon taxes or emission-trading systems) is based on internalizing externalities, causing a direct influence on organizational financial and budgeting management (Chen and Tseng, 2011). The theory is that ever-rising implied carbon prices under market-based mechanisms will induce innovation in low- or no-carbon technologies. With these additional long-term costs there is an assumed strategic commitment by organizations furthering support for technological and organizational innovations. But, in practice there is still a question on whether a small economy can provide the marginal benefits of investing in carbon abatement through these mechanisms (Ergas, 2012). In addition, the two major mechanisms of fixed price versus tradable permits may have very different managerial decision outcomes due to variations in risk and uncertainty associated with each mechanism (Chen and Tseng, 2011). Nevertheless, in some trading situations where the cap on the number tradable permits is very tight (a low cap) even market trading mechanisms may act as a tax after certain prices are set (Sarkis and Tamarkin, 2005).

The organizational implications of these government regulatory policies have been studied from a number of perspectives including the development of carbon accounting within organizations (Stechemesser and Guenther, 2012), determining when to invest in technology (Sarkis and Tamarkin, 2005), and the types of products and processes that organizations should integrate (Schaltegger and Csutora, 2012). In this study the focus is on how a carbon tax policy mechanism may influence SC decisions in a closed-loop system, causing organizations to rethink operational and network design considerations.

2.2. Green supply chain management and closed-loop supply chains

Additional field-focused contextual background is provided through an overview of GSCM and CLSC management research that relates to modeling environmental and carbon (greenhouse gas) emissions that informs some of our research. GSCM is by definition a way to improve the environmental performance and reduce the environmental burdens of SCs (Sarkis, 2003; Zhu and Sarkis, 2004). CLSC can benefit the environment but also cause additional environmental burdens. CLSC practices and activities can be environmentally beneficial from a product stewardship and extension of product/materials life perspective that help in mitigating resource

depletion and environmental footprints related to disposal of material and products (Meade et al., 2007). The activities and outcomes in CLSC[5] that are environmentally beneficial include reduction, reuse, reclamation of material, result in the reduction of wastes and resource depletion (El korchi and Millet, 2011). Although these resource depletion concerns can be met through CLSC operations, the current investigation turns to the environmental implications of these CLSC operations activities from a carbon emissions management perspective. That is, the additional CLSC (reverse logistics) activities for an organization can cause greater overall carbon emissions. The focus of our work and the review of the models below do not necessarily consider these environmental tradeoffs and burden shift. Yet, the model introduced in this study should be part of a systemic perspective in actual implementation to manage the concerns with environmental burden shifting (Jensen, 2012; Laurent et al., 2012). When linking life cycle analysis (LCA) to GSCM and CLSC, these types of shifting environmental burdens should be considered more explicitly.

Most of the past GSCM and CLSC studies have either been descriptive frameworks or conceptual models (Flapper et al., 2005). A recent review on GSCM literature shows that out of 308 papers published between 1990 and 2010 only 36 use quantitative, formal models (Seuring, 2013). Most of the published models with an explicit focus on broader GSCM address sustainability from the forward SC perspective using one of the five primary categories of modeling approach including analytical hierarchy process (AHP), game theoretic equilibrium, information theoretic approaches, LCA and multi-criteria decision making (MCDM). For example, Hsu and Hu (2009) apply the analytical hierarchy process for supplier selection in the context of hazardous substance management; Bai and Sarkis (2010) adopted rough set theory to develop green suppliers; and Sheu et al. (2005) show how sustainable practices can contribute to increased profit.

A CLSC incorporates a reverse logistics system designed to manage the flow of products or parts destined for reuse, recycle, remanufacturing or disposal (Bai and Sarkis, 2013; Carter and Ellram, 1998). The review of Fleischmann et al. (2000) shows that most of the analytical models published prior to 2000 address reuse and recycle aspects of reverse SC employing open-loop network systems. Fleischmann et al. (2000) identified only two studies that modeled CLSC systems including Jayaraman et al. (1999) and Krikke et al. (1999). After Fleischmann et al. (2000), at least two other reviews have been published investigating the characteristics of the reverse SC models (Pokharel and Mutha, 2009; Rubio et al., 2007). Both reviews conclude that the use of quantitative models for the analysis of CLSCs has increased substantially with the primary focus placed on the production planning and inventory management dimensions. Research on the analytical modeling and optimization of broader CLSCs is still limited. There is evidence suggesting that the optimization of CLSCs can be both environmentally sound and economically profitable (Guide et al., 2000; Jayaraman et al., 1999). Primary emphasis of these published models has been on business decisions, profitability, or cost minimization rather than the environmental aspects of remanufacturing and CLSCs. This research further exemplifies the secondary, but important, role of environmental issues as a side benefit of CLSCs.

[5] CLSC is heavily dependent on having a series of reverse logistics operations in addition to standard forward SC activities such as procurement, operations, manufacturing, and distribution. The major reverse logistics functional phases can include collection, separation, disassembly, compaction and outbound logistics (El korchi and Millet, 2011).

Economically focused analytical models for CLSCs have been the rule, with few explicitly considering environmental aspects. For example, Inderfurth (2005) developed a model of a CLSC that evaluates the optimal recovery and manufacturing policy in an uncertain environment. Teunter et al. (2008) investigated a situation where companies use shared and specialized resources for manufacturing and remanufacturing. Zuidwijk and Krikke (2008) studied the strategic questions of how much a company should invest in product design and production processes to process their returned products.

More recent models have started to more explicitly incorporate environmental factors. Various emergent measures such as Eco-indicator 99 and LCA have been integrated into bi-objective MILP models that consider the tradeoff between investment costs and environmental impact (Bojarski et al., 2009; Dehghanian and Mansour, 2009; Hugo and Pistikopoulos, 2005). These models consider strategic investment decisions such as technology and raw materials selection, but not necessarily production planning implications. Other MILPs have extended research in this area by considering the economic and environmental impact of decisions regarding the planning and design of an SC such as linking partners, supplier selection, determining facility capacities in each time period, and assigning manufacturing and distribution tasks to the network nodes (Chaabane et al., 2012; Diabat et al., 2013). Constraints include the demand and production amounts, and the environmental impact associated with each node using various general environmental indicators (Bojarski et al., 2009; Jotzo and Betz, 2009). The issue of facility location, a strategic level analysis, taking into consideration carbon emissions in a CLSC further emphasizes the practical and research importance of carbon sensitivity in the SC (Chaabane et al., 2012; Diabat et al., 2013). While these modeling efforts have focused on strategic–tactical level analysis, in this study we place our emphasis on the tactical–operational level analysis in a CLSC and develop an optimization model that is used to investigate cost and environmental influences of the a carbon pricing scheme on the tactical–operational planning contingencies.

3. Model development

The CLSC problem under investigation is shown in Fig. 1. In the forward network, multiple product types (i) are produced in different manufacturing plants (m) by travelling through a set of machine centers (g). Machine center g has its own production cost and carbon emission rate for processing one unit of product i. Finished products are then shipped to the end-users (e) either directly or via a set of warehouses (w). Product shipments may occur through different transport modes (k) with different prices and fuel efficiency rates (based on the type of vehicles used). In the reverse network, the end-of-life (EOL) products are gathered by the collection centers (c) where they are inspected and separated into recyclable and non-recyclable components. The non-recyclable components are destroyed at disposal centers (d). The recyclable components are sent to recycling centers (r) to be processed on a set of recycling machine centers (g'). Recycled raw materials are then shipped to the manufacturing plants for reuse in production line.

In terms of the modeling approach, a mathematical optimization technique is chosen over a simulation modeling technique for one primary reason. The problem under investigation can be formulated using MILP, in which both objective functions and modeling constraints are *linear* in form and all decision variable are either *continuous* or *binary* (0–1). A relatively large MILP model is solvable (for exact solution, not heuristically) using standard linear programming solvers such as CPLEX (Fahimnia, 2011; Fahimnia

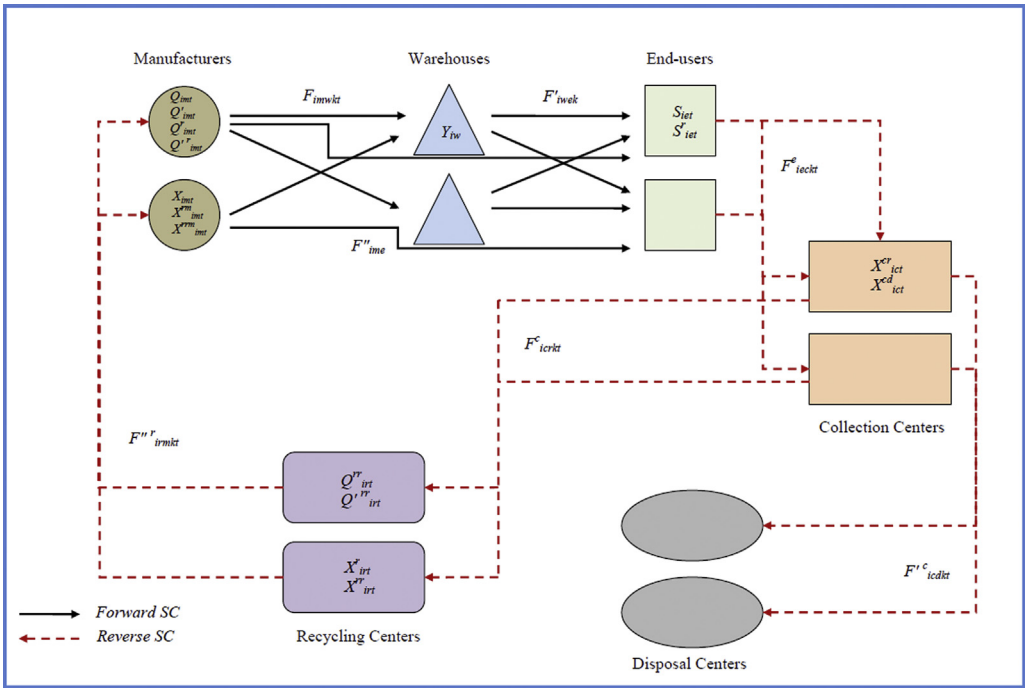


Fig. 1. Decision variables in the CLSC under investigation

et al., 2012). The proposed MILP model aims to determine the tactical–operational SC planning decisions, including production and distribution allocation strategies for the next planning horizon T (comprising t time periods) in such a way that the sum of manufacturing, transport and emission costs is minimized.

From the above problem statement the following indices/notations will be used for the purpose of mathematical modeling:

Product index	i	Set of products	I
Manufacturing plant index	m	Set of manufacturing plants	M
Machine center index	g	Set of machine centers	G
Warehouse index	w	Set of warehouses	W
Transport mode index	k	Set of transport modes	K
End-user index	e	Set of end-users	E
Time period index	t	Planning horizon	T
Collection center index	c	Set of collection centers	C
Recycling center index	r	Set of recycling centers	R
Recycling machine center index	g'	Set of recycling machine centers	G'
Disposal center index	d	Set of disposal centers	D

Assumptions in the proposed CLSC model include the following:

- Variety of products (i) to be produced is known.
- Number, location and capacity of plants (m) and warehouses (w) are known.
- Number and location of end-users (e) are known (i.e. customer zones or retailers).
- Number, location and capacity of collection centers (c), recycling centers (r) and disposal centers (d) are known.
- Demand for all product types is assumed to be known (deterministic) at all end-users for all periods of the next planning horizon.
- The forecasted demand for each product has to be satisfied, sooner or later, during the planning horizon. A penalty cost may be incurred if the demand for a certain product at one period is backordered. The backordered demand is to be satisfied in the subsequent periods before the end of the planning horizon.
- The quantity of *EOL* products returned to end-users at each period is known.
- A penalty/holding cost is incurred for uncollected *EOL* products in end-users.
- A product type can be supplied from more than one manufacturing plant, but the shipment of products and components between two manufacturing plants is not allowed.
- Production, distribution, recycling and distribution capacity limitations are known.
- Raw material and finished product inventory levels are known at the start and end of the planning horizon.
- End-users are the locations where products are delivered to the final customers with no holding capacity to store the products.
- Manufacturing and distribution air emissions are known for the processing of products and shipment of finished products from plants to end-users. The emission rates are determined according to the manufacturing technology adopted in manufacturing and transportation mode used in transport (e.g. older machines/trucks produce more emission).
- Cost of emission for holding/storing products (in plants, warehouses, collection centers, recycling centers and disposal centers) is assumed to be negligible when compared to the overall SC emission.

The input parameters and decision variables used in the proposed model are presented in Appendix 1. Using these notations, the MILP formulation of the proposed CLSC planning model with a unified objective function is presented in this section. For modeling purpose, we divide the proposed unified objective function into two dependent functions: cost function and emission function.

3.1. Formulation of cost function

The cost function is comprised of 14 elements (Eqs. (1)–(14)). Eq. (1) formulates the cost of opening and operating manufacturing plants (Term 1), warehouses (Term 2), collection centers (Term 3), recycling centers (Term 4), and disposal centers (Term 5).

$$Z_1 = \sum_t \sum_m o_{mt} G_{mt} + \sum_t \sum_w o'_{wt} G'_{wt} + \sum_t \sum_c o^c_{ct} G^c_{ct} + \sum_t \sum_r o^r_{rt} G^r_{rt} + \sum_t \sum_d o^d_{dt} G^d_{dt} \quad (1)$$

Eq. (2) formulates the second component of the cost function including manufacturing cost using virgin material in regular-time (Term 1) and (Term 2) overtime.

$$Z_2 = \sum_i \sum_m \sum_t \left[Q_{imt} \left(\sum_g p_{igmt} l_{igmt} + \omega_{imt} + \alpha_{imt} \right) \right] + \sum_i \sum_m \sum_t \left[Q'_{imt} \left(\sum_g p_{igmt} l'_{igmt} + \omega_{imt} + \beta_{imt} \right) \right] \quad (2)$$

Manufacturing cost using recycled material in regular-time (Term 1) and overtime (Term 2) is presented in Eq. (3).

$$Z_3 = \sum_i \sum_m \sum_t \left[Q^r_{imt} \left(\sum_g p_{igmt} l_{igmt} + \alpha_{imt} \right) \right] + \sum_i \sum_m \times \sum_t \left[Q'^r_{imt} \left(\sum_g p_{igmt} l'_{igmt} + \beta_{imt} \right) \right] \quad (3)$$

Cost of holding finished products in manufacturing plants (Term 1) and warehouses (Term 2) is given in Eq. (4).

$$Z_4 = \sum_i \sum_m \sum_t X_{imt} h_{imt} + \sum_i \sum_w \sum_t Y_{iwt} h'_{iwt} \quad (4)$$

Eq. (5) formulates the cost of holding virgin raw material and recycled raw material in manufacturing plants.

$$Z_5 = \sum_i \sum_m \sum_t (h^{rm}_{imt} X^{rm}_{imt} + h^{rrm}_{imt} X^{rrm}_{imt}) \quad (5)$$

Eq. (6) expresses the forward transportation cost for the shipment of finished products from plants to warehouses (Term 1), from warehouses to end-users (Term 2), and from plants to end-users (Term 3).

$$Z_6 = \sum_i \sum_m \sum_w \sum_k \sum_t F_{imwkt} \tau_{imwkt} + \sum_i \sum_w \sum_e \sum_k \times \sum_t F'_{iwekt} \tau'_{iwekt} + \sum_i \sum_m \sum_e \sum_k \sum_t F''_{imekt} \tau''_{imekt} \quad (6)$$

Cost of recycling in (1) regular-time and (2) overtime is presented in Eq. (7).

$$Z_7 = \sum_i \sum_r \sum_t \left[Q^{rr}_{irt} \left(\sum_g p^r_{ig'rt} l^r_{ig'rt} + \alpha'_{irt} \right) \right] + \sum_i \sum_r \sum_t \left[Q'^{rr}_{irt} \left(\sum_g p^r_{ig'rt} l'^r_{ig'rt} + \beta'_{irt} \right) \right] \quad (7)$$

Cost of collection/inspection/separation of *EOL* products at collection centers is calculated by Eq. (8).

$$Z_8 = \sum_i \sum_e \sum_c \sum_k \sum_t F^e_{ieckt} \pi^{ins}_{ic} \quad (8)$$

Cost of destroying the non-recyclable components at the disposal centers is presented in Eq. (9).

$$Z_9 = \sum_i \sum_c \sum_d \sum_k \sum_t F'^c_{icdkt} \pi^{dis}_{id} \quad (9)$$

Eq. (10) presents the cost of holding recyclable and disposing material in collection centers.

$$Z_{10} = \sum_i \sum_c \sum_t \left(h^{cr}_{ict} X^{cr}_{ict} + h^{cd}_{ict} X^{cd}_{ict} \right) \quad (10)$$

Cost of holding recyclable and recycled material in recycling centers is given in Eq. (11).

$$Z_{11} = \sum_i \sum_r \sum_t (h^r_{irt} X^r_{irt} + h^{rr}_{irt} X^{rr}_{irt}) \quad (11)$$

Eq. (12) formulates the cost of purchasing *EOL* products (Term 1) as well as the cost of transporting *EOL* products from end-users to collection centers (Term 1), transporting recyclable products from collection centers to recycling centers (Term 2), transporting disposing material from collection centers to disposal centers (Term 3), and transporting recycled material from recycling centers to manufacturing plants (Term 4).

$$Z_{12} = \sum_i \sum_e \sum_c \sum_k \sum_t F^e_{ieckt} (\tau^e_{ieckt} + \omega'_{iet}) + \sum_i \sum_c \sum_r \sum_k \sum_t F^c_{icrkt} \tau^c_{icrkt} + \sum_i \sum_c \sum_d \sum_k \sum_t F'^c_{icdkt} \tau'^c_{icdkt} + \sum_i \sum_r \sum_m \sum_k \sum_t F''^r_{irmkt} \tau^r_{irmkt} \quad (12)$$

Penalty cost of uncollected *EOL* products at the end-users is presented in Eq. (13).

$$Z_{13} = \sum_i \sum_e \sum_t S^r_{iet} pc_{iet} \quad (13)$$

Eq. (14) presents backordering cost of unsatisfied demand at end-users.

$$Z_{14} = \sum_i \sum_e \sum_t S_{iet} sc_{iet} \quad (14)$$

The cost function, presented in Eq. (15), is the sum of the 14 linear cost components presented in Eqs. (1)–(14).

$$Z = Z_1 + Z_2 + Z_3 + Z_4 + Z_5 + Z_6 + Z_7 + Z_8 + Z_9 + Z_{10} + Z_{11} + Z_{12} + Z_{13} + Z_{14} \quad (15)$$

3.2. Formulation of emission function

The emission function has four elements presented in Eqs. (16)–(19). The generated manufacturing carbon emission is presented in Eq. (16).

$$Z'_1 = \sum_i \sum_g \sum_m \sum_t (Q_{imt} + Q'_{imt} + Q^r_{imt} + Q'^r_{imt}) p_{igmt} c_{igmt} \quad (16)$$

Carbon emission at recycling centers is presented in Eq. (17).

$$Z'_2 = \sum_i \sum_{g'} \sum_r \sum_t (Q^{rr}_{irt} + Q'^{rr}_{irt}) p^r_{ig'rt} c^r_{ig'rt} \quad (17)$$

Forward transport pollution, expressed in Eq. (18), comprises of the generated emission for the shipment of finished products from plants to warehouses (Term 1), from warehouses to end-users (Term 2), and from plants directly to end-users (Term 3):

$$Z'_3 = \sum_i \sum_m \sum_w \sum_k \sum_t F_{imwkt} a_{imwkt} + \sum_i \sum_w \sum_e \sum_k \sum_t F'_{iwekt} a'_{iwekt} + \sum_i \sum_m \sum_e \sum_k \sum_t F''_{imekt} a''_{imekt} \quad (18)$$

Reverse transport pollution (Eq. (19)) comprises of carbon emission generated for the shipment of *EOL* products from end-users to collection centers (Term 1), recyclable products from collection centers to recycling centers (Term 2), disposing material from collection centers to disposal centers (Term 3), and recycled material from recycling centers to manufacturing plants (Term 4).

$$\begin{aligned} Z'_4 = & \sum_i \sum_e \sum_c \sum_k \sum_t F^e_{ieckt} a^e_{ieckt} \\ & + \sum_i \sum_c \sum_r \sum_k \sum_t F^c_{icrkt} a^c_{icrkt} \\ & + \sum_i \sum_c \sum_d \sum_k \sum_t F'^c_{icdkt} a'^c_{icdkt} \\ & + \sum_i \sum_r \sum_m \sum_k \sum_t F''^r_{irmkt} a^r_{irmkt} \end{aligned} \tag{19}$$

The emission function in Eq. (20) is formed by the summation of the four linear cost components presented in Eqs. (16)–(19):

$$Z' = Z'_1 + Z'_2 + Z'_3 + Z'_4 \tag{20}$$

The unified goal of the proposed optimization model is to minimize the value of Z in Eq. (21) in which the emission function is presented in terms of dollar carbon cost.

$$Z = Z + k\,Z' \tag{21}$$

3.3. Constraints

The proposed CLSC model is subject to the following set of linear constraints.

3.3.1. Constraints at manufacturing plants

Inventory balance of raw material in manufacturing plants:

$$Q_{imt} + Q'_{imt} + X^{rm}_{imt} - X^{rm}_{im(t-1)} \le \gamma_{imt} \quad \forall\, i,m,t \tag{22}$$

$$X^{rm}_{im0} = \eta^r_{im} \quad \& \quad X^{rm}_{imT} = \eta'^r_{im} \quad \forall\, i,m \tag{23}$$

Inventory balance of recycled raw material in manufacturing plants:

$$X^{rrm}_{imt} - X^{rrm}_{im(t-1)} = \sum_r \sum_k F''^r_{irmkt} - (Q^r_{imt} + Q'^r_{imt})\mu^{rp}_i \quad \forall\, i,m,t \tag{24}$$

$$X^{rrm}_{im0} = \eta^{rr}_{im} \quad \& \quad X^{rrm}_{imT} = \eta'^{rr}_{im} \quad \forall\, i,m \tag{25}$$

Production capacity constraint (machine center capacity limitation) for regular-time and overtime production in manufacturing plants:

$$\begin{aligned} & (Q_{imt} + Q^r_{imt})\, p_{igmt} \le \lambda_{igmt} \quad \& \\ & (Q'_{imt} + Q''^r_{imt}) p_{igmt} \le \lambda'_{igmt} \quad \forall\, i,g,m,t \end{aligned} \tag{26}$$

Holding capacity restriction at manufacturing plants:

$$X_{imt} \le hc_{imt} \quad \forall\, i,m,t \tag{27}$$

Inventory balance of products at plants:

$$\begin{aligned} & Q_{imt} + Q'_{imt} + Q^r_{imt} + Q''^r_{imt} - X_{imt} + X_{im(t-1)} \\ & = \sum_w \sum_k F_{imwkt} + \sum_e \sum_k F''_{imekt} \quad \forall\, i,m,t \end{aligned} \tag{28}$$

$$X_{im0} = \eta_{im} \quad \& \quad X_{imT} = \eta'_{im} \quad \forall\, i,m \tag{29}$$

3.3.2. Constraints at collection centers

Inventory balance of recyclable material of *EOL* product i in collection centers:

$$\begin{aligned} X^{cr}_{ict} - X^{cr}_{ic(t-1)} = & w^{EOL}_i \mu^{r\max}_i \sum_e \sum_k F^e_{ieckt} \\ & - \sum_r \sum_k F^c_{icrkt} \quad \forall\, i,c,t \end{aligned} \tag{30}$$

$$X^{cr}_{ic0} = \varphi^c_{ic} \quad \& \quad X^{cr}_{icT} = \varphi^{ct}_{ic} \quad \forall\, i,c \tag{31}$$

Inventory balance of disposed material of *EOL* product i in collection centers:

$$\begin{aligned} X^{cd}_{ict} - X^{cd}_{ic(t-1)} = & w^{EOL}_i (1 - \mu^{r\max}_i) \sum_e \sum_k F^e_{ieckt} \\ & - \sum_d \sum_k F'^c_{icdkt} \quad \forall\, i,c,t \end{aligned} \tag{32}$$

$$X^{cd}_{ic0} = \varphi'^c_{ic} \quad \& \quad X^{cd}_{icT} = \varphi'^{ct}_{ic} \quad \forall\, i,c \tag{33}$$

Collection centers capacity restriction:

$$\sum_e \sum_k F^e_{ieckt} \le \sigma^c_{ict} \quad \& \quad X^{cr}_{ict} + X^{cd}_{ict} \le hc^c_{ict} \quad \forall\, i,c,t \tag{34}$$

3.3.3. Constraints at recycling centers

Inventory balance of recyclable material in recycling centers:

$$X^r_{irt} - X^r_{ir(t-1)} = \sum_c \sum_k F^c_{icrkt} - (Q^{rr}_{irt} + Q'^{rr}_{irt})\mu^{rr}_i \quad \forall\, i,r,t \tag{35}$$

$$X^r_{ir0} = \varphi^r_{ir} \quad \& \quad X^r_{irT} = \varphi'^r_{ir} \quad \forall\, i,r \tag{36}$$

Inventory balance of recycled raw material in recycling centers:

$$X^{rr}_{irt} - X^{rr}_{ir(t-1)} = Q^{rr}_{irt} + Q'^{rr}_{irt} - \sum_m \sum_k F''^r_{irmkt} \quad \forall\, i,r,t \tag{37}$$

$$X^{rr}_{ir0} = \varphi^{rr}_{ir} \quad \& \quad X^{rr}_{irT} = \varphi'^{rr}_{ir} \quad \forall\, i,r \tag{38}$$

Production capacity constraint (machine center capacity limitation) for regular-time and overtime production in recycling centers:

$$Q^{rr}_{irt} p^r_{ig'rt} \le \lambda^r_{ig'rt} \quad \& \quad Q'^{rr}_{irt} p^r_{ig'rt} \le \lambda'^r_{ig'rt} \quad \forall\, i,g',r,t \tag{39}$$

Holding capacity constraint at recycling centers:

$$X^r_{irt} \le hc^r_{irt} \quad \& \quad X^{rr}_{irt} \le hc^{rr}_{irt} \quad \forall\, i,r,t \tag{40}$$

3.3.4. Constraints at disposal centers

Disposing capacity restriction:

$$\sum_c \sum_k \sum_t F'^c_{icdkt} \le \sigma^d_{id} \quad \forall\, i,d \tag{41}$$

3.3.5. Constraints at warehouses

Warehouse capacity restriction:

$$Y_{iwt} \le hc'_{iwt} \quad \forall\, i,w,t \tag{42}$$

Inventory balance at warehouses:

$$Y_{iwt} - Y_{iw(t-1)} = \sum_m \sum_k F_{imwkt} - \sum_e \sum_k F'_{iwekt} \quad \forall\, i,w,t \tag{43}$$

$$Y_{im0} = \varphi_{iw} \quad \& \quad Y_{iwT} = \varphi'_{iw} \quad \forall\, i,w \tag{44}$$

3.3.6. Distribution constraints

Distribution capacity limits at plants:

$$F_{imwkt} \le \varepsilon^{\max}_{imwkt} \quad \forall\, i,m,w,k,t \tag{45}$$

$$F''_{imekt} \le \delta^{\max}_{imekt} \quad \forall\, i,m,e,k,t \tag{46}$$

Distribution capacity constraint at warehouses:

$$F'_{iwekt} \le \theta^{\max}_{iwekt} \quad \forall\, i,w,e,k,t \tag{47}$$

Distribution capacity constraint at customer zones:

$$F^e_{ieckt} \le \varepsilon^{e\max}_{ieckt} \quad \forall\, i,e,c,k,t \tag{48}$$

Distribution capacity constraint at collection centers:

$$F^c_{icrkt} \le \varepsilon^{r\max}_{icrkt} \quad \forall\, i,c,r,\ k,t \tag{49}$$

$$F'^c_{icdkt} \le \varepsilon^{d\max}_{icdkt} \quad \forall\, i,c,d,\ k,t \tag{50}$$

Distribution capacity constraint at recycling centers:

$$F''^r_{irmkt} \le \theta^{r\max}_{irmkt} \quad \forall\, i,r,m,\ k,t \tag{51}$$

3.3.7. Demand satisfaction constraints

Demand satisfaction constraint: The total amount of production for every product at all plants must meet the forecasted demand for that product at the end of planning horizon unless there is inventory in manufacturing plants and warehouses at $t = 0$ and $t = T$.

$$\sum_m \sum_t (Q_{imt} + Q'_{imt} + Q^r_{imt} + Q'^r_{imt}) = \sum_e \sum_t d_{iet} + \sum_m \eta'_{im} - \sum_m \eta_{im} + \sum_w \varphi'_{iw} - \sum_w \varphi_{iw} \quad \forall\, i \tag{52}$$

Maximum allowed shortage at end-users:

$$S_{iet} \le s^{\max}_{iet} \quad \forall\, i,e,t \tag{53}$$

Maximum allowed for uncollected *EOL* products:

$$s^r_{iet} \le s^{r\max}_{iet} \quad \forall\, i,e,t \tag{54}$$

Inventory balance at end-users:

$$\sum_w \sum_k F'_{iwekt} + \sum_m \sum_k F''_{imekt} = d_{iet} - S_{iet} + S_{ie(t-1)} \quad \forall\, i,e,t \tag{55}$$

Inventory balance at end-users for *EOL* products:

$$\sum_k \sum_c F^e_{ieckt} = d'_{iet} - S^r_{iet} + S^r_{ie(t-1)} \quad \forall\, i,e,t \tag{56}$$

3.3.8. Carbon emission constraints

Carbon emission constraint in manufacturing plants:

$$\sum_i \sum_g p_{igmt} c_{igmt} (Q_{imt} + Q'_{imt} + Q^r_{imt} + Q'^r_{imt}) \le \frac{c^{\max}_{mt}}{1000} \quad \forall\, m,t \tag{57}$$

Carbon emission constraint in recycling centers:

$$\sum_i \sum_{g'} p^r_{ig'rt} c^r_{ig'rt} (Q^{rr}_{irt} + Q'^{rr}_{irt}) \le \frac{c^{r\max}_{rt}}{1000} \quad \forall\, r,t \tag{58}$$

Carbon emission constraint in transport:

$$\sum_i \sum_m \sum_w \sum_k F_{imwkt} a_{imwkt} + \sum_i \sum_w \sum_e \sum_k F'_{iwekt} a'_{iwekt} + \sum_i \sum_m \sum_e \sum_k F''_{imekt} a''_{imekt} + \sum_i \sum_e \sum_c \sum_k F^e_{ieckt}\, a^e_{ieckt} + \sum_i \sum_c \sum_r \sum_k F^c_{icrkt}\, a^c_{icrkt} + \sum_i \sum_c \sum_d \sum_k F'^c_{icdkt} a'^c_{icdkt} + \sum_i \sum_r \sum_m \sum_k F''^r_{irmkt} a^r_{irmkt} \le \frac{a^{\max}_t}{1000} \quad \forall\, j,t \tag{59}$$

3.3.9. Restrictions on decision variables

$$0 \le Q_{imt} \le G_{mt} L \quad \& \quad 0 \le Q'_{imt} \le G_{mt} L \quad \forall\, i,m,t \tag{60}$$

$$0 \le Q^r_{imt} \le G_{mt} L \quad \& \quad 0 \le Q'^r_{imt} \le G_{mt} L \quad \forall\, i,m,t \tag{61}$$

$$0 \le Q^{rr}_{irt} \le G^r_{rt} L \quad \& \quad 0 \le Q'^{rr}_{irt} \le G^r_{rt} L \quad \forall\, i,r,t \tag{62}$$

$$0 \le F_{imwkt} \le G_{mt} L \ \& \ 0 \le F_{imwkt} \le G'_{wt} L \quad \forall\, i,m,w,k,t \tag{63}$$

$$0 \le F'_{iwekt} \le G'_{wt} L \quad \forall\, i,w,e,k,t \tag{64}$$

$$0 \le F''_{imekt} \le G_{mt} L \quad \forall\, i,m,e,k,t \tag{65}$$

$$0 \le F^e_{ieckt} \le G^c_{ct} L \quad \forall\, i,e,c,k,t \tag{66}$$

$$0 \le F^c_{icrkt} \le G^c_{ct} L \quad \& \quad 0 \le F^c_{icrkt} \le G^r_{rt} L \quad \forall\, i,c,r,k,t \tag{67}$$

$$0 \le F'^c_{icdkt} \le G^c_{ct} L \quad \& \quad 0 \le F'^c_{icdkt} \le G^d_{dt} L \quad \forall\, i,c,d,k,t \tag{68}$$

$$0 \le F''^r_{irmkt} \le G^r_{rt} L \quad \& \quad 0 \le F''^r_{irmkt} \le G_{mt} L \quad \forall\, i,r,m,k,t \tag{69}$$

$$0 \le X_{imt} \quad \& \quad 0 \le X^{rm}_{imt} \quad \& \quad 0 \le X^{rrm}_{imt} \quad \forall\, i,m,t \tag{70}$$

$$0 \le Y_{iwt} \quad \forall\, i,w,t \tag{71}$$

$$0 \le X^{cr}_{ict} \quad \& \quad 0 \le X^{cd}_{ict} \quad \forall\, i,c,t \tag{72}$$

$$0 \le X^r_{irt} \quad \& \quad 0 \le X^{rr}_{irt} \quad \forall\, i,r,t \tag{73}$$

$$0 \le S_{iet} \quad \& \quad 0 \le S^r_{iet} \quad \forall\, i,e,t \tag{74}$$

4. Model implementation

TexF is involved in the provision of fibrous material used in car seats, carriers, sofas, dining chairs, filling material, and seat covers. The wide range of textile products manufactured at TexF can meet the needs of a wide spectrum of industries including automotive and furniture. For about a decade, the high volume of damaged and returned goods at TexF resulted in large amount of product waste. In 2011, the company decided to convert to a CLSC by recycling its damaged and *EOL* products returning them to the production line as some sort of recycled raw material. Such activities and processes as color sorting, fiber blending, cleaning, weaving, knitting, compressing, and shredding are a part of the reverse network at the collection and recycling centers. Through developing a CLSC not only could TexF gain some cost reduction benefits, but it has also been a way to build a long-term reputation in the sector. TexF is now pondering how the manufacturing and distribution strategies of its CLSC are affected with the recently introduced carbon pricing scheme in Australia. This became the primary motivation for conducting this research and developing a unified CLSC model that explicitly considers the reduction of carbon pollution in both forward and reverse courses.

TexF has two manufacturers in South Australia and Queensland ($M = 2$). Plant 1 with older machinery is cheaper to run but generates more carbon emission. Plant 2 generates less emission, but is costlier to produce items. Each plant is equipped with six machine centers ($G = 6$) involved in the production of more than 10 product types. Here, we target the SC planning of four main product families at TexF including fiber car seats, baby carriers, sofas, and dining chairs ($I = 4$). Each product may or may not need to travel through all six machine centers. Finished products satisfy the market demand at five customer locations ($E = 5$) through three warehouses ($W = 3$). The reverse network has five collection centers ($C = 5$), two disposal centers ($D = 2$), and three recycling centers ($R = 3$) each equipped with four machine centers (G'=4). Three transport modes are available in all the routes ($K = 3$). Transport mode 1 is the cheapest among the three while produces more emission per unit transportation. Transport mode 2 has an intermediate position and transport mode 3 is the most expensive but greenest option. The planning horizon is defined to be one year consisting of 12 one-month time periods ($T = 12$).

The authors were initially approached by the case company representatives seeking advice on the impact of carbon pricing on the TexF's CLSC. The data collection for this case study was a long multi-step process. Forecasted demand data, backordering allowances and costs, as well as the historical product return data at the collection centers were obtained from the sales and marketing department for the financial year 2010–2011. For the same planning period, the SC management unit at TexF provided the manufacturing, recycling and transportation capacities and costs as well as the inventory costs and holding capacities in plants, warehouses, collection centers and recycling centers. Carbon emission data was available for every machine in manufacturing plants and recycling centers. Transportation emission rates were estimated from the average values obtained from two third-party logistics providers.

The proposed model in Section 3 was coded using AMPL modeling language with CPLEX as the core optimization solver. The numerical results are converted into Excel spreadsheets for analysis purposes. Using the real data collected from TexF's CLSC operations, the model was run for four hypothetical carbon prices ($0, $20, $30 and $40 per ton of emission) in two scenarios: *forward SC scenario (FS)* and *CLSC scenario (CLS)*. These prices represent a realistic range given market-based policy instrument schemes in other nations and regions of the world (Chapple et al., 2013; Sarkis and Tamarkin, 2005). FS disregards the reverse SC operations and therefore uses virgin materials *only* to produce the products. CLS represents the status quo in TexF in which a reverse SC is adopted to reduce the use of virgin materials in production lines. Tables 1 and 2 present the numerical results for FS and CLS respectively. The numerical results include the model outputs for the four different carbon prices and include the overall SC cost for each carbon price as well as the values of the disaggregated objective function components (Eqs. (15) and (20)). It should be noted that the cost function in Eq. (15) represents the SC cost excluding the associated emission costs. Emission costs are derived from multiplying emission function by the associated carbon cost (k).

In these tables, '*Prod cost*' is the cost of production and inventory holding in manufacturing plants. '*Forw Dist*' shows the cost of forward transport from plants to end-users as well as the cost of inventory holding in warehouses. For the reverse SC, '*Coll/Rec/Disp*' is the cost of collection, inspection, separation, recycling and disposal as well as the inventory holding costs in collection centers, recycling centers and disposal centers. '*Trans*' shows the accumulative cost for the shipment of *EOL* products from end-users back to the manufacturing plants. Analogues abbreviations are used for the components of the emission function presenting the amount of emission generated for corresponding activities in the cost function.

5. Discussion

Various insights can be gained from the variations in carbon pricing. Carbon emissions pricing may be dependent on a variety of forces in the market place or through regulatory policies. Depending on the policies set forth by the Australian government or strategies set forth by industries and even individual organizations, the pricing of carbon emission can change quite drastically. We vary the prices over a small and realistic range given the history of these types of market-based policy instrument schemes in other nations and regions of the world (Chapple et al., 2013; Sarkis and Tamarkin, 2005). The resulting reactions to these pricing changes in terms of optimal costs and emissions based on two types of SC scenarios and decisions set the stage for the optimization analysis. we design and model two situations, one with a reverse network (CLS) and one with only a forward SC (FS).

Intuitively, as carbon prices rise, the operational costs increase while emission rates decrease. This situation assumes similar rates in demand and manufacturing of products. In a closed-loop environment complexity in the analysis increases due to potential cost savings and/or cost increases from additional steps in the process. Thus, our investigation at this time is focused on what costs and emissions results occur from an SC planning model with and without closing the loop. It is difficult to make initial conjectures about the emissions and economic results for both of these cases. The only conjecture that can be made is that we can expect decreases in optimal design carbon emissions when carbon costs (prices) increase.

Table 1
Numerical results for Scenario 1 (FS).

Carbon price ($/ton)	Overall SC cost ($) from Eq. (21)	Cost function (Z)		Emission function (Z') * k	
		Forw SC costs ($)		Forw emiss costs ($)	
		Prod	Forw dist	Prod	Forw dist
0	8,622,713	6,055,013	2,567,700	0	0
20	8,727,515	6,061,017	2,568,118	73,810	24,570
40	8,822,990	6,064,231	2,575,057	138,314	45,388
60	8,914,450	6,066,557	2,591,613	195,500	60,780

Table 2
Numerical results for Scenario 2 (CLS).

Carbon price ($/ton)	Overall SC cost ($) from Eq. (21)	Cost function (Z)				Emission function (Z') * k			
		Forw SC costs ($)		Rev SC costs ($)		Forw emiss costs ($)		Rev emiss costs ($)	
		Prod	Forw dist	Coll/Rec/Disp	Trans	Prod	Forw dist	Coll/Rec/Disp	Trans
0	8,381,213	4,700,960	2,568,330	826,668	285,255	0	0	0	0
20	8,514,598	4,706,097	2,566,176	831,693	285,629	73,435	24,904	20,941	5723
40	8,637,980	4,709,235	2,571,482	837,201	287,830	137,122	46,210	38,358	10,542
60	8,752,325	4,710,015	2,586,355	841,504	290,351	194,820	62,580	52,580	14,120

The organization will be interested in knowing the optimal carbon emissions, especially if carbon prices can be assigned to all activities managed by the organization. If a larger decrease in carbon emissions occurs with the CLSC, organizations like TexF may be willing to incorporate the CLSC to help gain larger credit amounts when available. The ultimate decision facing the organization is what CLSC operations to adopt in presence of a carbon pricing scheme. It noteworthy that even though there may be greater overall carbon emissions from additional reverse SC operations, CLSC practices can be environmentally beneficial from a product stewardship and extension of products/materials life perspective that help in mitigating resource depletion and environmental footprints related to disposal of material and products. Here, we only focus our CLSC analysis on the implications from a carbon emissions management perspective leaving the resource depletion discussion and shift of environmental burden (Finkbeiner, 2009; Jensen, 2012) out of the area of our research focus.

We begin by varying the carbon prices in an exclusively forward SC scenario (FS) and a CLSC scenario (CLS) with ranges from 0 to 60 dollars per ton of carbon emissions. Fig. 2 shows the optimal carbon emissions trend for FS at different carbon prices and Fig. 3 illustrates the abatement potential achieved when CLSC is in place. The analysis for the FS considers the emissions from the production and forward distribution stages of TexF's operations. Here, transport emissions represent less than 25% of the overall carbon emission and the remaining 75% belongs to the manufacturing emissions. This result may not be the case for many other types of organizations, but it is the case for TexF. The general breakdown of operational emissions sources can provide valuable information for organizations that can target where reductions are more feasible and where the largest 'bang-for-the-buck' exists.

Comparing the differentials of optimal solutions for the FS in Fig. 2, as carbon price increases from $0 to $60 per ton, the total optimal production emissions decreases by 18.6%. The percentage (and absolute value) decreases at a slower marginal rate as the price increases. For example, the decrease in emissions is 7.8%, or 315 tons, when the carbon price increases from $0 to $20 per ton, and only 6.3% (233 tons) for the carbon price increasing from $20 to $40 per ton.

The total carbon emission in CLS is primarily generated at the production stage and only to a minor percentage by the underlying transport flows. In the reverse network, the recycling processes generate considerably higher emissions than the related transport operations. For CLSC, the contribution of manufacturing emission is less than 60% in the overall carbon pollution while reverse operations contribute up to 22%.

Interestingly, for the distribution activities of the FS, the resulting optimal emissions reduction rate over a varying price range shows an increasing marginal decrease in emissions as price increases. Although the total optimal distribution emission decreases by 18.2%, a very similar rate to the production stage of the FS, the optimal emission reductions are smaller at the lower price increases. That is, the rate of change of carbon emissions for the FS distribution stage is only 0.6%, down by only 8 tons, when the price

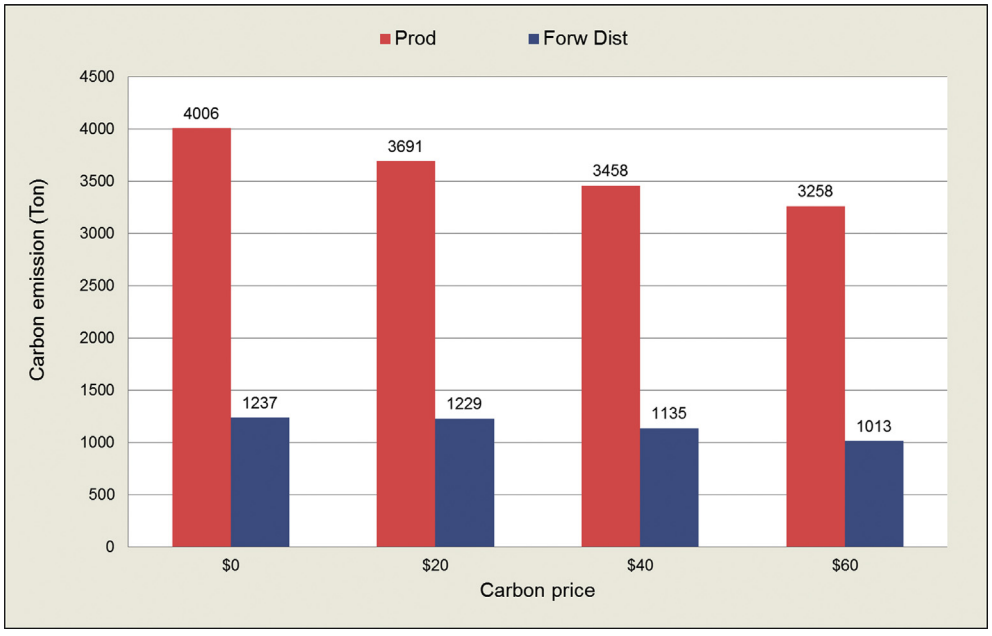


Fig. 2. Generated carbon emission at various carbon prices for Scenario 1 (FS).

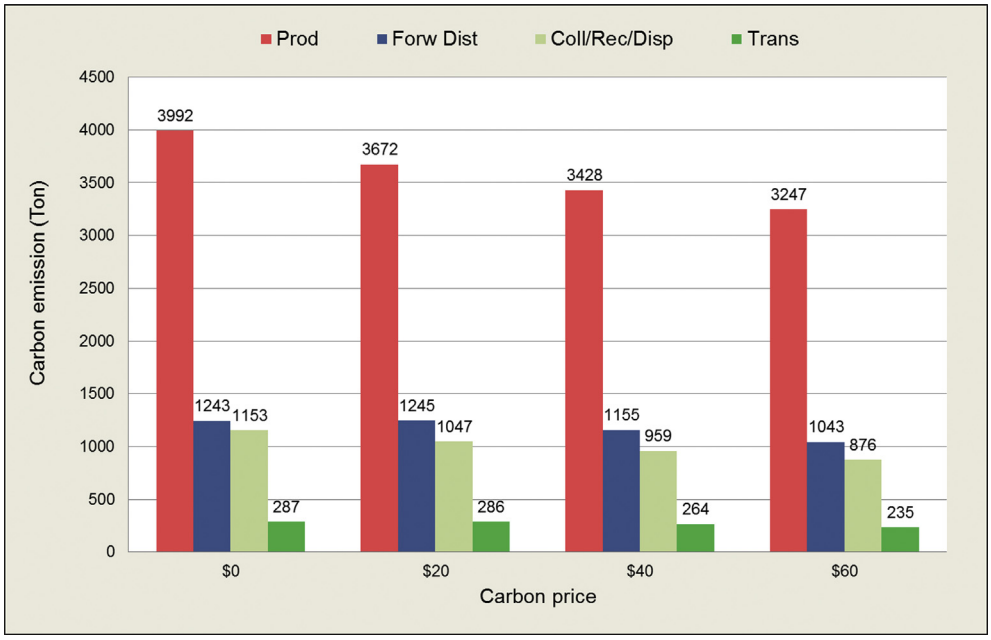


Fig. 3. Generated carbon emission at various carbon prices for Scenario 2 (CLS).

changes from $0 to $20 per ton. The carbon emission decreases by 10.7% if price increases from $40 to $60 per ton. This increasing marginal emission reduction results is opposite of the production stage's trend.

The initial implication from these numerical results is now evident. If an organization is seeking to invest in technologies or practices to help decrease their carbon emissions, it would be more worthwhile, from an absolute and percentage reduction perspective to focus on production processes and technology at lower level carbon pricing. If the carbon emission market pricing gets increases to a larger value of over $40/ton, the shift in investigating carbon emission savings through new technologies and practices should occur in the transportation and distribution functions.

Operating a CLSC can positively affect the 'green image' of TexF through its resource consumption reduction efforts. Alternatively, managing the direct operations of a CLSC can also generate considerable additional carbon emissions that appear on the organization's operational books. That is, what would have been considered a Scope 3 emission (indirect SC emission), in the use of recycled material, would now become a Scope 1 emission (direct operations) that needs to be managed by TexF in a CLSC model (see Hoffmann and Busch (2008) for a discussion of Scope for Carbon Auditing). It has been shown that the embedded carbon emissions of products and materials from recycling operations tend to be more carbon efficient than virgin material procurement (Edwards et al., 2010) where significant emission is generated in raw material extraction and the associated processing activities.

CLSC environmental advantages are primarily related to reducing resource consumption. While a more systemic, bigger picture, analysis shows that the global emissions can be reduced through recycling, for TexF, running a CLS means incurring about 20% more carbon pollution when compared to FS. The global emission abatement that TexF's CLS offers need to be investigated with the inclusion of virgin material carbon emission values. Overall, this situation with FS performing better from a carbon emissions perspective can be discouraging for establishing CLSCs when a carbon pricing scheme is in place. Focusing only on one environmental metric (e.g. carbon emissions) can cause environmental burden shift that organizations need to be concerned about, and thus more complete life cycle analyses with multiple dimensions is recommended to arrive at a more accurate environmental burden portfolio with and without CLSC activities (Finkbeiner, 2009; Jensen, 2012).

Once the scheme is converted into a carbon trading mechanism in the near future, mandatory caps on emissions will limit the amount of carbon pollution unless extra permits are available to purchase. One policy implication that can be drawn from these findings is that carbon costs incurred via reverse SC operations may need to be *subsidized* to a large extent by the governments in carbon pricing regulations. Alternatively, additional carbon costs and pricing for an FS system that only uses virgin materials with greater carbon footprints may be a policy to even the playing field.

For the closing the loop scenario (CLS), Figs. 4 and 5 show the overall SC cost increase (Fig. 4) and emission improvement (Fig. 5) for three carbon prices ($20, $40, $60 per ton) compared to the baseline of $0 carbon price. A cost analysis of the optimal results for variations in carbon prices shows for the CLS, in Fig. 4, the optimal solution costs are generally increasing as carbon prices increase. This result is not surprising and is expected due to the additional carbon costs incurred. Interestingly, at the $20/ton carbon price, the distribution stage of the SC optimal cost solution decreases slightly (Fig. 4) along with the corresponding slight emissions generation increase (Fig. 5).

Our analysis, here, is focused on discovering the relationship between the rate at which costs are increased and the corresponding emission improvement rate. Fig. 5 indicates with carbon price rising from $0 to $20 per ton of emission, transport practices show only minor adaptation, while forward manufacturing and reverse operations show more significant emission improvements. However, the emission improvement in reverse operations (Fig. 5) is supported by the significant cost increase in these facilities (Fig. 4). This observation indicates that for smaller carbon prices, investing on forward manufacturing can be the most economical way of emission saving. It should however be noted that the absolute manufacturing cost is about 5.7 times greater than the cost of the reverse operations (Coll/Rec/Disp), while the emission generation is only 3.5 times as big. This implies the manufacturing processes are inherently greener than reverse operations.

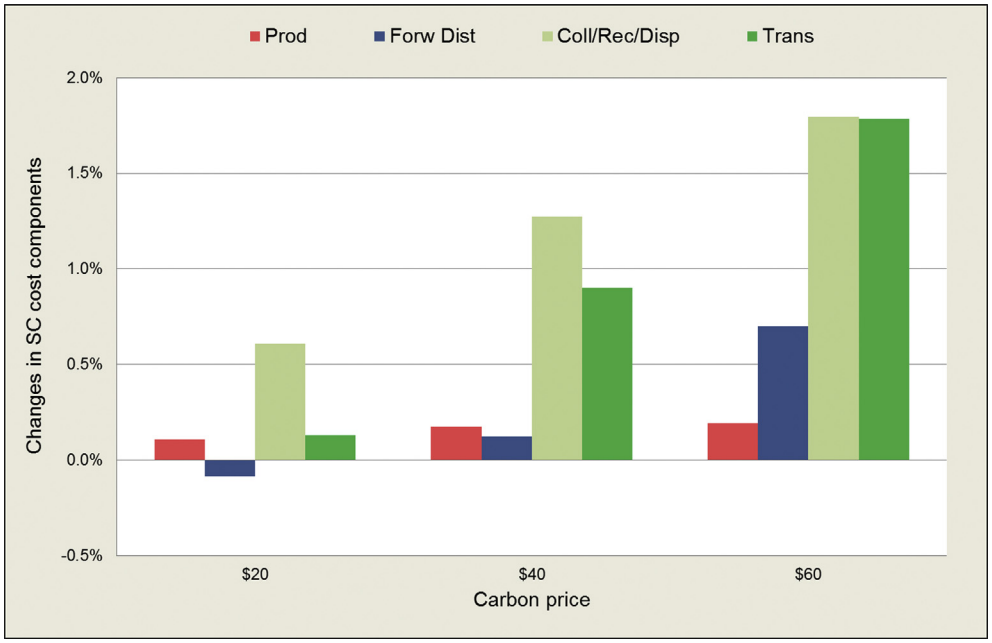


Fig. 4. Changes in SC cost components for CLS (compared to the baseline of $0 carbon price).

While there is a very consistent transport cost increase when carbon price rises from $0 to $60 (Fig. 4), for carbon prices greater than $20 significant improvements can be observed in transport emissions (Fig. 5). A clear managerial implication from this observation is that for larger carbon prices the primary focus in a CLSC should be given to the use of greener transport modes and technologies, while smaller carbon prices may require the adaptation of more carbon-efficient production and recycling systems.

A direct comparative analysis of the overall costs and emission improvement trends for FS and CLS are shown in Figs. 6 and 7 respectively. Fig. 6 shows that the absolute cost values for FS are greater than those of CLS over the range of carbon prices. This is obviously due to the savings in the decreased use of virgin material at CLS and hence the lower cost of raw material procurement. Fig. 7, on the other hand, shows that the absolute carbon emission values are greater in CLS because of the carbon emissions generated via the reverse SC activities. In fact, Fig. 7 fails to illustrate the emission savings that could be obtained through the reduced extraction, production, and transportation of virgin materials. In other words, part of the overall emissions that is now being generated by TexF as a Scope 1 emission (direct emission) was traditionally a Scope 3 emission managed by the raw material suppliers.

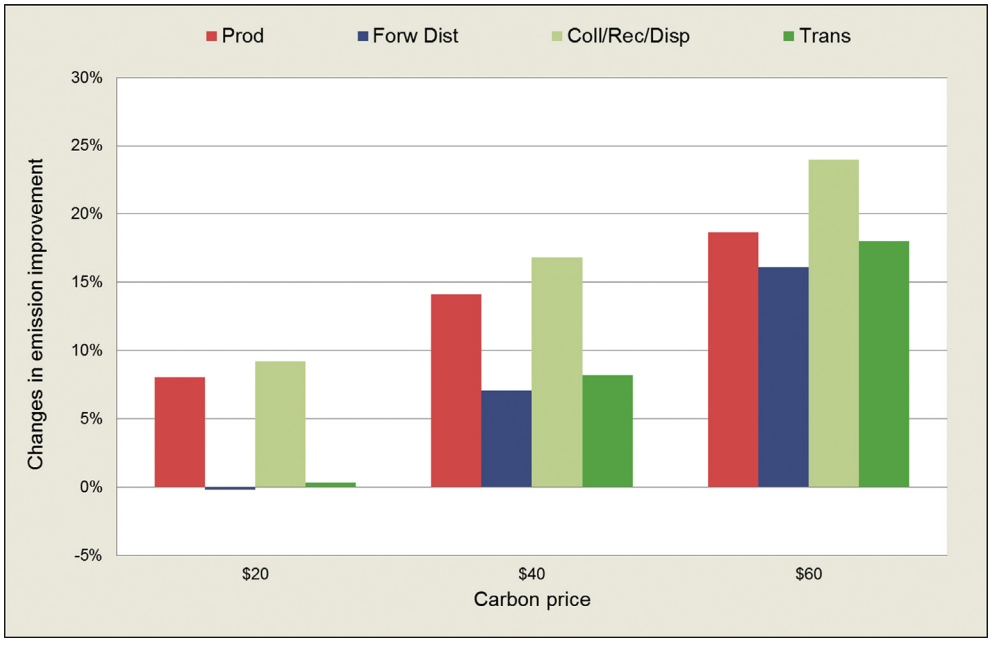


Fig. 5. Changes in carbon savings for CLS (compared to the baseline of $0 carbon price).

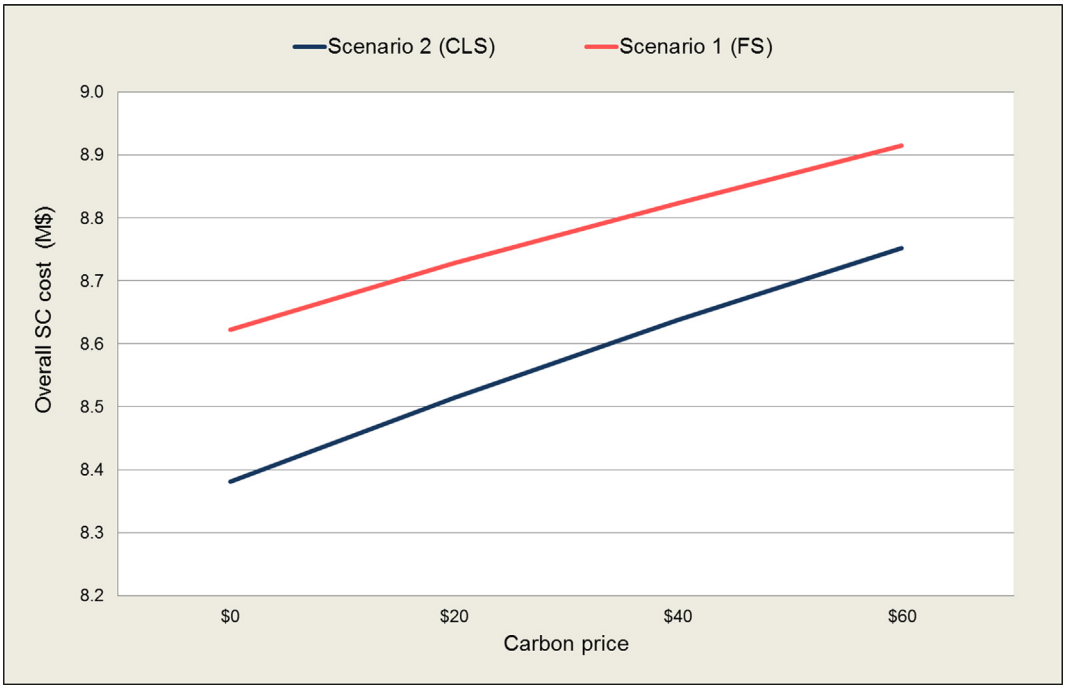


Fig. 6. Trends in SC cost increase for FS and CLS.

Another observation is that the overall cost of CLS seems to be increasing at a faster rate compared to FS (evidenced by the clear convergence of the two lines in Fig. 6); while the emissions generations are improved at a closer rates (the slighter convergence rate of the two lines Fig. 7). These two latter observations indicate how the establishment of a CLSC can be discouraging when the carbon price gets to be larger. Overall, the need to subsidize the carbon costs incurred via the reverse network becomes more visible at larger carbon prices.

Fig. 8 compares the trend in the reduction of carbon emissions for FS and CLS with the trend in overall cost increase in FS and CLS respectively. The values shown are incremental that compare the increased costs and emission reductions to those at the preceding carbon price. While the incremental increase in the optimal SC costs taper off as the carbon price increases from $20 to $60, the incremental change in carbon reduction benefits initially rise then level off for the same carbon price range, especially for the CLS. The tradeoff is evident, whereby organizations have greater emissions reductions for a given carbon price increases. At the larger carbon prices (closer to $60 per ton), the incremental emission reductions in the CLS are similar to those of the FS. In fact, the progression seems to show the CLS incremental emissions reductions will be less than FS incremental emissions reductions. But, both systems still have substantial savings in emissions.

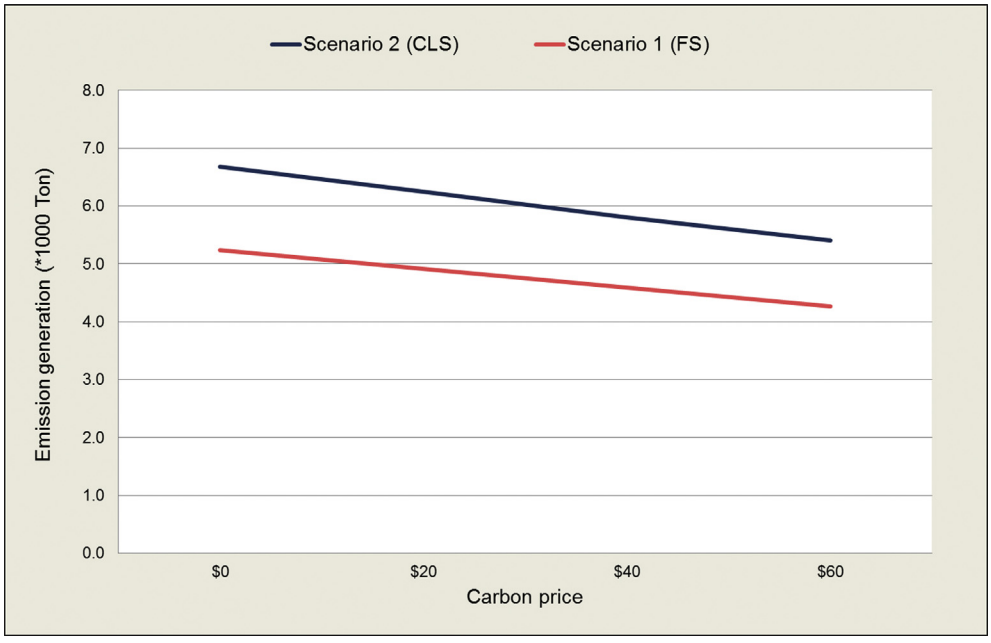


Fig. 7. Trends in emission generation for FS and CLS.

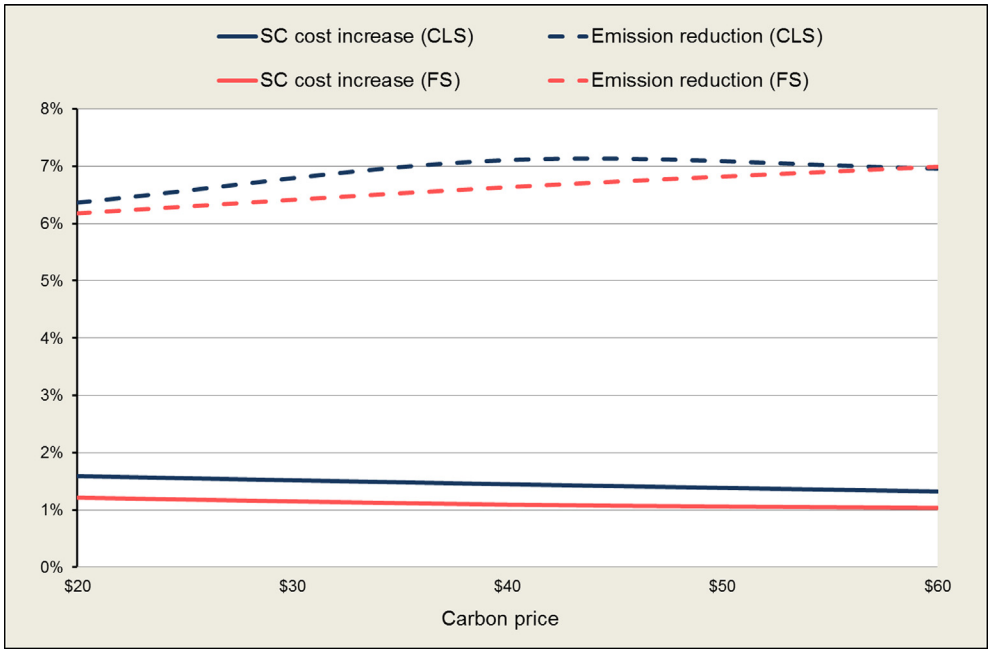


Fig. 8. Incremental changes in SC cost and emission generation for FS and CLS.

The trends in SC cost increase and emission reduction shows that higher carbon prices can lead to less incremental SC cost increases but greater reduction in emission generation. This finding supports the basic reasoning by having carbon prices increase (if policy makers are to tax at these rates) and further support investments in emissions reduction technology.

From Fig. 8, the most significant incremental improvement in emissions generation reduction happens to be in the CLS and in the carbon price range of $20–$40. From a policy perspective, assuming all companies fall within the same characteristics parametric pricing, taxing at this rate or managing the market price to remain in this range will give the largest incremental benefit for CLS. If the FS is used, the incremental reductions seem to be continuously improving as the carbon prices increase. From an organizational perspective, companies can gain competitive advantages by focusing on FS at higher rates (outsourcing reverse operations) if they wish to show the higher emissions reductions.

6. Conclusions

The decoupling of environmental degradation from economic growth requires that institutions consider the joint improvement of both dimensions when planning their SCs. This balance decision-making has been a concern at the national and regional levels. This paper takes this decoupling and 'win–win' proposition to the SC level. We developed an optimization model that seeks to balance an SC carbon footprint (emission reduction goal) and tactical economic goals. The impacts of carbon pricing on both a standard forward SC and a CLSC were evaluated. Parametric analysis was conducted over various carbon prices utilizing real case settings. This study is practically valuable since Australia, where the case company is located, is currently implementing a carbon pricing market. The development of such optimization models is particularly valuable for organizations, researchers and policy makers.

We arrive at some important findings from our initial analyses. First, corporations need to evaluate the impact of carbon pricing on their SC performance with various implications related to which operations and level of '*scope*' to consider when designing and determining their carbon footprints. This scoping is important especially when such impacts are significant and companies may need to establish comprehensive strategic approaches in response to the introduced carbon pricing schemes.

To promote SC decarbonization, governments may need to provide subsidies on carbon costs incurred via reverse SC operations. As could be seen in our results, the optimal designs may involve various improvement or deterioration in some SC functions. Identifying these areas is critical for planning purposes because government subsidization and industry focus may be placed on SC functions where more improvements can be gained. One issue is that expanding the scope of control to include CLSC activities may actually be detrimental to an organization's carbon footprint, but other environmental benefits should not be ignored. Subsidizing some of the reverse SC processes would make the CLSC operations more economically viable especially at larger carbon prices. Incorporating additional carbon costs for upstream virgin material emissions can be included in future models to illustrate that CLSLs are not necessarily worse.

What we have described are only one set of implications based on some initial analysis from the model development and execution. Further research could investigate the influence of significant factors that may have impacts on the SC performance such as the rate at which products are returned, product recyclability and the capacity of reverse operations. Also, the cost of virgin material may vary with changes in the emission prices (our model disregarded the carbon pollution generated through the extraction and delivery of raw material at the supplier side). A more complete analysis of a CLSC needs to include all three Scopes of emission generation. We believe that the proposed model in this paper is flexible enough to include such information, but such actual data may not be easily available to all companies, as in our case study. These are possible extensions, but also identify some of the limitations in our technique.

What we have studied is an important and critical issue facing industry, government, communities and other stakeholders. The study builds the foundation for optimization of carbon in CLSC environment. The utilization of optimization

techniques to investigate environmental issues is a step in the direction for development of more compassionate operations (Sarkis, 2012b) that include social and environmental sustainability issues.

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Appendix 1. Model parameters and decision variables

Parameters

d_{iet}	Forecasted demand for product i in e at t (units)
d'_{iet}	Quantity of returned *EOL* product i in e at t (units)
w_i^{EOL}	Weight of *EOL* product i (kg)
μ_i^{rmax}	Maximum recyclable fraction of *EOL* product i (%)
μ_i^{rr}	Consumption fraction of recyclable material for producing a unit of recycled raw material for product i (%)
μ_i^{rp}	Unit consumption of recycled material for producing a unit of product i (kg/unit)
o_{mt}	Fixed costs of opening and operating m at t ($)
o'_{wt}	Fixed costs of opening and operating w at t ($)
o_{ct}^{c}	Fixed costs of opening and operating c at t ($)
o_{rt}^{r}	Fixed costs of opening and operating r at t ($)
o_{dt}^{d}	Fixed costs of opening and operating d at t ($)
h_{imt}	Unit holding cost for product i in m at t ($/unit)
h'_{iwt}	Unit holding cost for product i in w at t ($/unit)
h_{imt}^{rm}	Unit holding cost for virgin raw material i in m at t ($/unit)
h_{imt}^{rrm}	Unit holding cost for recycled raw material i in m at t ($/kg)
h_{ict}^{cr}	Unit holding cost for recyclable material of *EOL* product i in c at t ($/kg)
h_{ict}^{cd}	Unit holding cost for disposed material of *EOL* product i in c at t ($/kg)
h_{irt}^{r}	Unit holding cost for recyclable material of *EOL* product i in r at t ($/kg)
h_{irt}^{rr}	Unit holding cost for recycled raw material i in r at t ($/kg)
hc_{imt}	Maximum holding capacity for product i in m at t (units)
hc'_{iwt}	Maximum holding capacity for product i in w at t (units)
hc_{ict}^{c}	Maximum holding capacity for *EOL* product i (both recyclable and disposed material) in c at t (units)
hc_{irt}^{r}	Maximum holding capacity in r for recyclable material of *EOL* product i at t (kg)
hc_{irt}^{rr}	Maximum holding capacity in r for recycled raw material of *EOL* product i at t (kg)
σ_{ict}^{c}	Maximum inspection/separation capacity for *EOL* product i in c at t (units)
σ_{id}^{d}	Maximum disposing capacity for disposed material of *EOL* product i in d (kg)
π_{id}^{dis}	Cost of destroying a unit of disposed material of *EOL* product i in d ($/kg)
π_{ic}^{ins}	Cost of inspection/separation of a unit of *EOL* product i in c ($/unit)
p_{igmt}	Processing time to produce a unit of product i on g in m at t (h/unit)
$p_{ig'rt}^{r}$	Processing time to recycle a unit of recyclable material of *EOL* product i on g' in r at t (h/kg)
l_{igmt}	Labour/hour cost for regular-time production of product i on g in m at t ($/h)
l'_{igmt}	Labour/hour cost for overtime production of product i on g in m at t ($/h)
$l_{ig'rt}^{r}$	Labour/hour cost for regular-time production of recycled raw material i on g' in r at t ($/h)
$l_{ig'rt}^{rr}$	Labour/hour cost for overtime production of recycled raw material i on g' in r at t ($/h)
ω_{imt}	Cost of raw material required for producing a unit of i in m at t ($/unit)
ω'_{iet}	Cost of purchasing a returned *EOL* product i in e at t ($/unit)
α_{imt}	Variable overhead cost of regular-time production of i in m at t ($/unit)
α'_{irt}	Variable overhead cost of regular-time recycling for a unit of recycled raw material i in r at t ($/kg)

(*continued*)

β_{imt}	Variable overhead cost of overtime production of product i in m at t ($/unit)
β'_{irt}	Variable overhead cost of overtime recycling for a unit of recycled raw material i in r at t ($/kg)
sc_{iet}	Unit backordering (shortage) cost for product i in e at t ($/unit)
s_{iet}^{max}	Maximum backordering permitted for product i in e at t ($/unit)
pc_{iet}	Penalty/inventory cost for uncollected *EOL* product i in e at t ($/unit)
s_{iet}^{rmax}	Maximum allowed capacity for uncollected *EOL* product i in e at t (units)
λ_{igmt}	Capacity hours for regular-time production of product i on g in m at t (h)
$\lambda_{ig'rt}^{r}$	Capacity hours for regular-time recycling of *EOL* product i on g' in r at t (h)
λ'_{igmt}	Capacity hours for overtime production of product i on g in m at t (h)
$\lambda'^{r}_{ig'rt}$	Capacity hours for overtime recycling of *EOL* product i on g' in r at t (h)
γ_{imt}	Capacity units of raw material supply for product i in m at t (units)
τ_{imwkt}	Unit transportation cost of product i from m to w through k at t ($/unit)
τ'_{iwekt}	Unit transportation cost of product i from w to e through k at t ($/unit)
τ''_{imekt}	Unit transportation cost of product i from m to e through k at t ($/unit)
τ_{ieckt}^{e}	Unit transportation cost of *EOL* product i from e to c through k at t ($/unit)
τ_{icrkt}^{c}	Unit transportation cost of recyclable material of *EOL* product i from c to r through k at t ($/kg)
τ'^{c}_{icdkt}	Unit transportation cost of disposed material of *EOL* product i from c to d through k at t ($/kg)
τ_{irmkt}^{r}	Unit transportation cost of recycled raw material of *EOL* product i from r to m through k at t ($/kg)
$\varepsilon_{imwkt}^{max}$	Maximum allowed distribution of product i from m to w through k at t (units)
θ_{iwekt}^{max}	Maximum allowed distribution of product i from w to e through k at t (units)
δ_{imekt}^{max}	Maximum allowed distribution of product i from m to e through k at t (units)
$\varepsilon_{ieckt}^{emax}$	Maximum allowed distribution of *EOL* product i from e to c through k at t (units)
$\varepsilon_{icrkt}^{rmax}$	Maximum allowed distribution of recyclable material of *EOL* product i from c to r through k at t (kg)
$\varepsilon_{icdkt}^{dmax}$	Maximum allowed distribution of disposing material of *EOL* product i from c to d through k at t (kg)
θ_{irmkt}^{rmax}	Maximum allowed distribution of recycled raw material of *EOL* product i from r to m through k at t (kg)
η_{im}	Inventory level of product i in m at the start of planning horizon (units)
η'_{im}	Inventory level of product i in m at the end of planning horizon (units)
η_{im}^{r}	Inventory level of raw material i in m at the start of the planning horizon (kg)
η'^{r}_{im}	Inventory level of raw material i in m at the end of the planning horizon (kg)
η_{im}^{rr}	Inventory level of recycled raw material i in m at the start of planning horizon (kg)
η'^{rr}_{im}	Inventory level of recycled raw material i in m at the end of planning horizon (kg)
φ_{iw}	Inventory level of product i in w at the start of planning horizon (units)
φ'_{iw}	Inventory level of product i in w at the end of planning horizon (units)
φ_{ic}^{c}	Inventory level of recyclable material of *EOL* product i in c at the start of planning horizon (kg)
φ_{ic}^{ct}	Inventory level of recyclable material of *EOL* product i in c at the end of planning horizon (kg)
φ'^{c}_{ic}	Inventory level of disposed material of *EOL* product i in c at the start of planning horizon (kg)
φ'^{ct}_{ic}	Inventory level of disposed material of *EOL* product i in c at the end of planning horizon (kg)
φ_{ir}^{r}	Inventory level of recyclable material of *EOL* product i in r at the start of planning horizon (kg)
φ'^{r}_{ir}	Inventory level of recyclable material of *EOL* product i in r at the end of planning horizon (kg)

(*continued on next page*)

(*continued*)

φ_{ir}^{rr}	Inventory level of recycled raw material of *EOL* product i in r at the start of planning horizon (kg)
$\varphi_{ir}^{'rr}$	Inventory level of recycled raw material of *EOL* product i in r at the end of planning horizon (kg)
c_{igmt}	Estimated carbon emission to produce a unit of product i on g in m at t (kg/h)
$c_{mt}^{\max}$	Maximum allowed carbon emission in m at t (ton)
$c_{ig'rt}^{r}$	Estimated carbon emission to produce a unit of recycled material from *EOL* product i on g' in r at t (kg/h)
$c_{rt}^{r\max}$	Maximum allowed carbon emission in r at t (ton)
a_{imwkt}	Estimated carbon emission for the shipment of product i from m to w through k at t (kg/unit)
$a_{iwekt}^{'}$	Estimated carbon emission for the shipment of product i from w to e through k at t (kg/unit)
$a_{imekt}^{''}$	Estimated carbon emission for the shipment of product i from m to e through k at t (kg/unit)
a_{ieckt}^{e}	Estimated carbon emission for the shipment of *EOL* product i from e to c through k at t (kg/unit)
a_{icrkt}^{c}	Estimated carbon emission for the shipment of recyclable material of *EOL* product i from c to r through k at t (kg/kg)
$a_{icdkt}^{'c}$	Estimated carbon emission for the shipment of disposed material of *EOL* product i from c to d through k at t (kg/kg)
a_{irmkt}^{r}	Estimated carbon emission for the shipment of recycled raw material of *EOL* product i from r to m through k at t (kg/kg)
$a_t^{\max}$	Maximum allowed transport emission at t (ton)
k	Carbon cost ($/kg)
L	'Big M' standing for a large number

Decision variables

Continuous variables:

Q_{imt}	Quantity of i produced in regular-time in m at t (units)
$Q_{imt}^{'}$	Quantity of i produced in overtime in m at t (units)
Q_{imt}^{r}	Quantity of i produced from recycled raw material in regular-time in m at t (units)
$Q_{imt}^{'r}$	Quantity of i produced from recycled raw material in overtime in m at t (units)
Q_{irt}^{rr}	Quantity of recycled raw material i produced in regular-time in r at t (kg)
$Q_{irt}^{'rr}$	Quantity of recycled raw material i produced in overtime in r at t (kg)
F_{imwkt}	Quantity of i shipped from m to w through k during t (units)
$F_{iwekt}^{'}$	Quantity of i shipped from w to e through k during t (units)
$F_{imekt}^{''}$	Quantity of i shipped directly from m to e through k during t (units)
F_{ieckt}^{e}	Quantity of *EOL* product i shipped from e to c through k during t (units)
F_{icrkt}^{c}	Quantity of recyclable material of *EOL* product i shipped from c to r through k during t (kg)
$F_{icdkt}^{'c}$	Quantity of disposed material i shipped from c to d through k during t (kg)
$F_{irmkt}^{''r}$	Quantity of recycled raw material i shipped from r to m through k during t (kg)
X_{imt}	Inventory amount of i in m at the end of t (units)
X_{imt}^{rm}	Inventory amount of raw material i in m at the end of t (kg)
X_{imt}^{rrm}	Inventory amount of recycled raw material i in m at the end of t (kg)
Y_{iwt}	Inventory amount of i in w at the end of t (units)
X_{ict}^{cr}	Inventory amount of recyclable material of *EOL* product i in c at the end of t (kg)
X_{ict}^{cd}	Inventory amount of disposing material of *EOL* product i in c at the end of t (kg)
X_{irt}^{r}	Inventory amount of recyclable material i in r at the end of t (kg)
X_{irt}^{rr}	Inventory amount of recycled raw material i in r at the end of t (kg)
S_{iet}^{r}	Quantity of uncollected *EOL* product i in e at the end of t (units)
S_{iet}	Quantity of i backordered in e at the end of t (units)

Binary variables:

$$G_{mt} = \begin{cases} 1, & \text{If } m \text{ operates in } t \\ 0, & \text{therwise} \end{cases}$$

$$G'_{wt} = \begin{cases} 1, & \text{If } w \text{ is open in } t \\ 0, & \text{Otherwise} \end{cases}$$

$$Gc_{ct} = \begin{cases} 1, & \text{If } c \text{ is open in } t \\ 0, & \text{Otherwise} \end{cases}$$

$$Gr_{rt} = \begin{cases} 1, & \text{If } r \text{ operates in } t \\ 0, & \text{Otherwise} \end{cases}$$

$$Gd_{dt} = \begin{cases} 1, & \text{If } d \text{ is open in } t \\ 0, & \text{Otherwise} \end{cases}$$

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